



[Segment Tree] $\Rightarrow$ Hard

question ?

② $\Rightarrow$

①

0	1	2	3	4	5	6	7
3	8	27	12	4	19	17	9
3	11	38	50	54	73	90	99

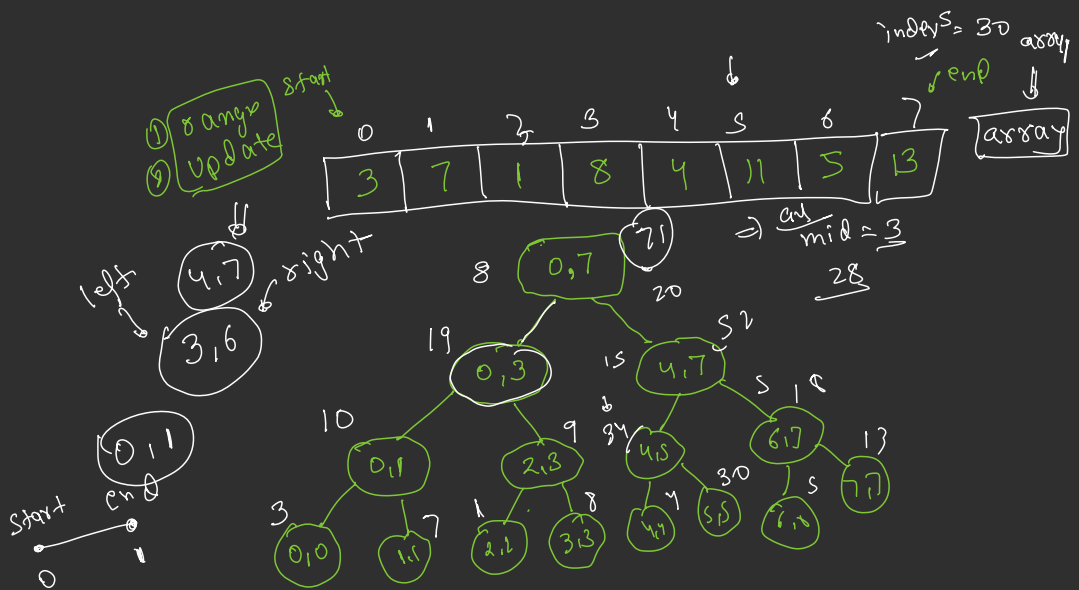
② $O(n)$

sum
range-over $\rightarrow [3, 6]$

$\rightarrow [1, 5]$

$\rightarrow [3, 5]$

① $\rightarrow$ range over
② $\rightarrow$ updation
 $N/2$
 $N/2$
 N
 $O(n^2)$



return 0

if (end < left || right < start)
return 0;

if (left <= start & end <= right)
return answer;

① ⇒ Out of range
② ⇒ Completely range
③ ⇒ Partial range

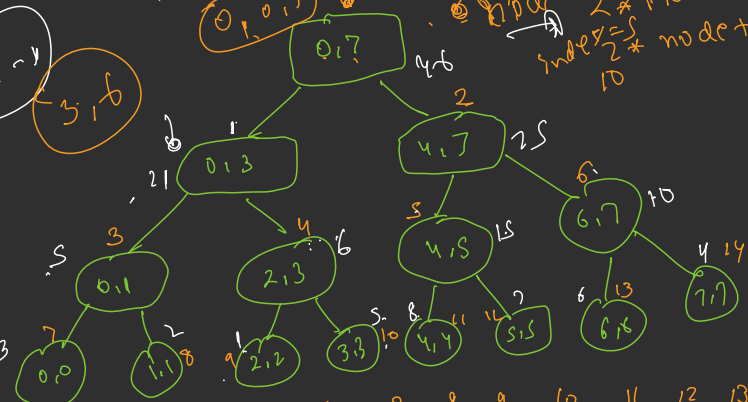
left right
3, 6
- -

0	1	2	3	4	5	6	7
3	2	1	5	8	7	6	4

$\Rightarrow 2n-1$
3, 6

0, 1, 0, 7

node
index
 $2 * \text{node} + 1$
 $2 * \text{node} + 2$
10



Array

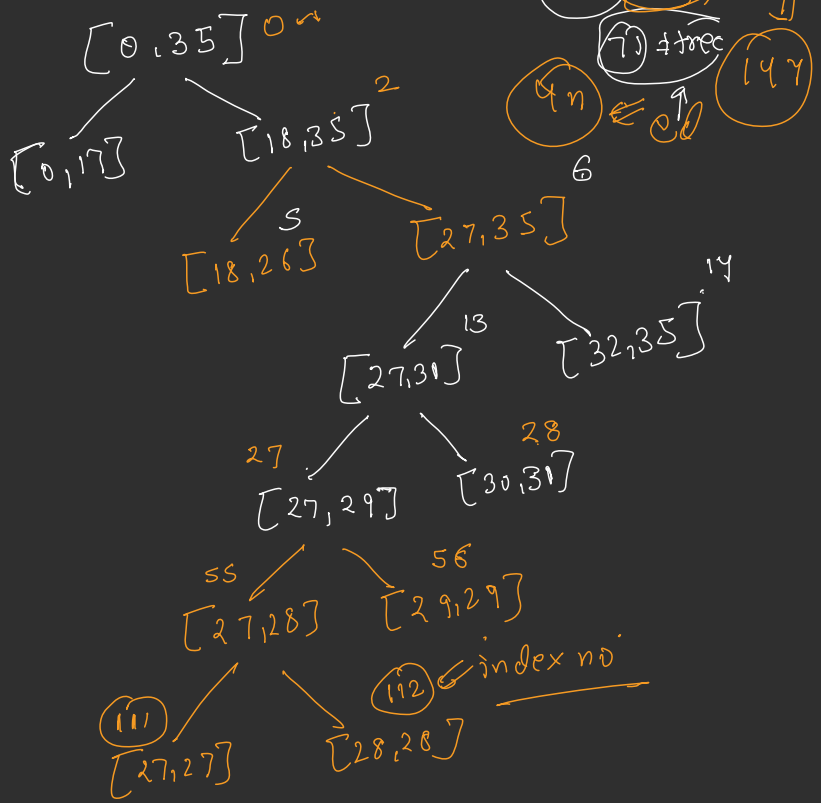
Tree width

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
40	11	25	5	6	15	10	3	2	1	5	8	7	6	4

node
Tree

$4n$

Page Number



0	1	2	3	4	5	6	7
10	7	1	3	2	5	9	4

~~10~~ ~~7~~ ~~1~~ ~~3~~ ~~2~~ ~~5~~ ~~9~~ ~~4~~



single vP

② update rule

(2, 6, 8)

(0, 3)

0	1	2	3	4	5	6	7
7	2	3	4	11	6	5	1

⇒ binary lazy propagation

OCW



lazy-array ✓